Harsh Shroff

AI/ML ENGINEER – GENERATIVE & AGENTIC SYSTEMS

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PROFESSIONAL SUMMARY

AI engineer building production generative and agentic systems: multi-agent orchestration (LangGraph), evaluation-gated RAG, and on-device vision-language deployment on NVIDIA Jetson. At UMBC's CARDS lab, contributed to a multimodal RGB/thermal/audio system deployed on a Boston Dynamics Spot robot under U.S. Army Research Lab funding. Shipped a compliance RAG platform on AWS at VITG. M.S. in Data Science; three peer-reviewed or indexed publications; AWS ML certified.

TECHNICAL SKILLS

Generative AI: LLMs, RAG (Chunking, Embeddings, Hybrid Search, pgvector/ChromaDB), Multimodal/VLM Inference, AWS Bedrock, Claude, GPT, Gemini

Agentic AI: LangGraph, Multi-Agent Orchestration, Tool Routing (MCP), LLM-as-Judge, Self-Correcting Evaluation Loops, PydanticAI

Computer Vision & Edge: YOLOv8, OpenCV, DINOv2, CLIP, ResNet, NVIDIA Jetson, TensorRT, Boston Dynamics Spot SDK

Speech: STT (Vosk, Whisper), TTS (Piper, EdgeTTS), End-to-End Voice Pipelines

Infrastructure: Python, FastAPI, Flask, SQL, PostgreSQL, vLLM, CUDA, Quantization (LoRA/QLoRA, INT4 GGUF), Docker, GitHub Actions, MLflow, AWS (Bedrock, EC2, Lambda, S3), Terraform

PROFESSIONAL EXPERIENCE

Applied AI Research Engineer **Mar 2023 – Aug 2026**

UMBC Center for Real-time Distributed Sensing and Autonomy

- **Contributed to a multimodal AI system** (RGB, thermal, audio, and VLM sensor fusion) deployed on a Boston Dynamics Spot robot under U.S. Army Research Lab funding. Co-authored the SPIE 2026 paper describing it.
- **Designed multi-agent orchestration systems** (LangGraph) with tool routing and dynamic escalation logic for autonomous reasoning in disconnected, zero-trust field environments, exposed through REST APIs.
- **Built evaluation infrastructure** (LLM-as-Judge) to catch hallucinations and low-confidence model outputs before they reached decision-makers.
- **Deployed INT4-quantized open-weight LLMs** (Llama 3.1 via Ollama and vLLM) and vision models (DINOv2, TensorRT/CUDA) on NVIDIA Jetson hardware alongside Boston Dynamics Spot SDK integrations; fine-tuned open-weight models with LoRA/QLoRA for intent classification, gated by LLM-as-Judge evaluation before hardware deployment.

AI Engineer **Jun 2024 – Jun 2026**

VITG Corp., Halethorpe, MD

- **Designed and shipped a production RAG platform on AWS** (pgvector semantic search, Claude 3.5 via AWS Bedrock) for compliance-critical queries, with audit logging built in from the start.
- **Set up evaluation and delivery pipelines** (GitHub Actions, MLflow) to track latency and answer quality across deployments.
- **Prototyped an MCP-style tool-routing layer** (an LLM classifier chooses the backend instead of a hardcoded mapping) and a Bedrock-backed resume-scoring Slack bot (Llama 3 8B on AWS Lambda), both deployed with Terraform.

Data Scientist – Computer Vision ML **Aug 2023 – Dec 2023**

The Conservation Fund, Shepherdstown, WV

- **Built computer-vision models** (YOLOv8, OpenCV) for fish-fillet quality analysis on edge hardware; co-authored the resulting peer-reviewed paper (*Elsevier JAFR*, 2024).

PROJECTS

MARS: Multi-Agent Research System (GitHub) (Live)

LangGraph, 11-Agent Orchestration, MCP, LLM-as-Judge

- Orchestrated 11 specialized LLM agents in a LangGraph state machine to produce citation-grounded research reports, with a self-correcting QC retry loop and parallel critic agents; shipped a standalone MCP server for external clients (Claude Desktop, Cursor).

– Measured an 87% first-pass QC rate (13/15 runs) and 0% citation-tag referential errors (tags point to real chunks, not a semantic-faithfulness measure) on a preliminary 15-run evaluation; the README states these limits openly.

VEST: Offline-First Edge Tactical AI (GitHub)
NVIDIA Jetson Orin, CUDA, Vision-Language Model, Multimodal RAG, STT/TTS

– Built a real-time edge system on NVIDIA Jetson Orin fusing live camera vision, offline speech-to-text (Vosk), an on-device vision-language model (Qwen3-VL via Ollama), and multimodal RAG (CLIP + ChromaDB) – about 4.1 s spoken query to spoken response on an AGX Orin 64GB (README measurement).
– Designed fallback chains with verbal acknowledgment for degraded conditions (camera offline, model timeout, low battery or thermal limits) instead of silent failure, and ran a dedicated latency-profiling and optimization pass.

Silicon Oracle: Production Agent Platform (GitHub) (Live)
Python, Flask, PostgreSQL, Automated Testing, CI/CD, Security Policy

– Live agent platform where every planned tool call passes a scoped, auditable permission layer before execution, with a keyword-routing fallback when the planning model is unavailable.
– Automated test suite, CI/CD pipeline, and a documented security policy; 70 commits, solo-designed and maintained.

Bio-Oracle: Neuro-Symbolic Agentic Screening Pipeline (GitHub)
Python, PyTorch (CUDA), Cellpose, PydanticAI, Docker

– Built a drug-screening pipeline pairing a Cellpose CUDA segmentation model with a PydanticAI agent that calls statistical outlier-detection tools instead of reasoning from raw pixels; 98% outlier precision on Taxol-induced actin changes in the Broad Institute BBBC021 screen.
– Containerized for headless batch runs (Docker, arm64) with MLflow and Arize Phoenix observability.

vLLM Inference Benchmark: A Study in Self-Correction (GitHub)
vLLM, PyTorch, CUDA, GPU Profiling

– Benchmarked vLLM continuous batching against a naive HuggingFace generation loop; found my first result was confounded by unseeded sampling variance, fixed the methodology, and re-verified before reporting.
– Defensible result: 1.6–2.2x higher throughput at batch sizes of 4 and up on matched-token prompts, with GPU memory profiling of vLLM's upfront KV-cache reservation.

AirbnbLens: Multi-Modal Recommendation Engine (GitHub)
FastAPI, PyTorch, ResNet152, ChromaDB, Next.js

– FastAPI backend serving photo- and text-based style search over 62,000+ listings across 8 cities, using ResNet152 image embeddings in a ChromaDB vector index.

EDUCATION

Master of Science, Data Science **Aug 2022 – May 2024**
University of Maryland Baltimore County (UMBC) — ***GPA: 3.8/4.0***

Bachelor of Engineering, Electronics & Communication **Aug 2018 – May 2022**
Gujarat Technological University (VGEC)

AWARDS

- **1st Place, ITU/TDRA International Virtual Hackathon** (Climate Change category, Jan 2022), among 1,319 participants; organized by the UAE TDRA and the ITU under the WSIS Forum

CERTIFICATIONS & PUBLICATIONS

- **AWS Machine Learning Associate** – Active through Apr 2028
- **Gurunathan, T.S., Kundurthi, M.H., Shroff, H.**, et al. (2026). "Multimodal AI for dismounted soldier-robot teaming and context-aware threat response." *Proc. SPIE 14043*, 1404307.
- **Ranjan, R., Shroff, H.**, et al. (2024). "FilletCam AI: A handheld tool for precise fillet color profiling of Atlantic salmon and rainbow trout." *Journal of Agriculture and Food Research.*
- **Trivedi, K., Shroff, H.** (2021). "Mosquito identification using ML on embedded systems." *IEEE ITU Kaleidoscope.*